

AIM: Methods of Strings

Steps for execution:

String Character (charAt()):

1. Open **Eclipse** or any Java IDE.
2. Create a new Java class named **StringCharacter**.
3. Write the charAt() program.
4. Save the file as **StringCharacter.java**.
5. Compile the program.
6. Run the program.

7. Output:

```
String: Hello Java  
Character at index 6: J
```

Execution Steps for String Concatenation:

1. Open Eclipse or any Java IDE.
2. Create a new Java project.
3. Create a Java class named StringConcat.
4. Write the string concatenation program using the concat() method.
5. Save the program.
6. Compile the program.
7. Run the program.

Output:

First String: Sai

Second String: Pujitha

Concatenated String: Sai Pujitha

Execution Steps for contains() Method:

1. Open **Eclipse** or any Java IDE.
2. Create a new Java class named **StringContains**.
3. Write the contains() program and save the file.
4. Compile the program.
5. Run the program.

6. The program checks whether the given text is present in the string.
7. If the text is present, it returns **true**; otherwise, it returns **false**.

8. Output:

```
String: Hello Java  
Contains Java: true
```

Execution Steps for endsWith() Method:

1. Write the Java program using the endsWith() method.
2. Save the file as StringEndsWith.java.
3. Open Command Prompt / Terminal.
4. Go to the folder where the Java file is saved.
5. If there are no errors, run the program.
6. The program checks whether the string ends with the specified text.
7. The result is displayed as **true** or **false**.

8. Output:

```
String: Hello Java  
Ends with Java: true
```

Execution Steps for String equals():

1. The program compares the two strings Open Eclipse, VS Code, or any Java IDE.
2. Create a new Java file named StringEquals.java.
3. Write the String equals() program in the file.
4. Save the program.
5. Compile the program.
6. using the equals() method.
7. If both strings contain the same text, it displays **true**; otherwise, it displays **false**.

Output:

First String: Pujitha

Second String: Pujitha

Strings are equal: true

Steps for execution for: string
equals, indexOf, isEmpty, lastIndexOf, length, replace, startswith, substring, tolower, toupper, trim:

1. Create the Java Program

Open Eclipse / VS Code / Notepad and create a Java file.

2. Write the Program

Write the Java program containing the following String methods:

- `indexOf()`
- `isEmpty()`
- `lastIndexOf()`
- `length()`
- `replace()`
- `startsWith()`
- `substring()`
- `toLowerCase()`
- `toUpperCase()`
- `trim()`

3. Save the Program

Save the file with the same name as the public class.

Example:

`StringMethods.java`

4. Compile the Program

Open the Command Prompt/Terminal and go to the folder containing the Java file.

```
javac StringMethods.java
```

If there are no errors, the program is successfully compiled.

5. Run the Program

```
java StringMethods
```

6. View the Output

The results of all the String methods will be displayed on the screen.

7. Verify the Output

Check whether the output correctly shows:

String: Sai Pujitha

Index of P: 4

String is empty: true
String: Hello Java
Last index of a: 9
String: Good Morning
Length of string: 12
Original String: Hello Java
After Replace: Hello World
String: Hello Java
Starts with Hello: true
Original String: Hello Java
Substring: Hello
Original String: HELLO WORLD
Lowercase String: hello world
Original String: My Collection
Uppercase String: MY COLLECTION
Original String: Hello Java
After trim: Hello Java

Program:

```
package corejava;  
  
public class StringCharacter {  
    public static void main(String[] args) {  
        String str = "Hello Java";  
        char result = str.charAt(6);  
  
        System.out.println("String: " + str);  
  
        System.out.println("Character at index 6: " + result);  
    }  
}
```

```
}  
  
}  
  
package corejava;  
  
public class StringConcat {  
  
    public static void main(String[] args) {  
  
        String str1 = "Sai";  
  
        String str2 = "Pujitha";  
  
        String result = str1.concat(" ").concat(str2);  
  
        System.out.println("First String: " + str1);  
  
        System.out.println("Second String: " + str2);  
  
        System.out.println("Concatenated String: " + result);  
  
    }  
  
}
```

```
package corejava;  
  
public class StringContains {  
  
    public static void main(String[] args) {  
  
        String str = "Hello Java";  
  
        boolean result = str.contains("Java");  
  
        System.out.println("String: " + str);  
  
        System.out.println("Contains Java: " + result);  
  
    }  
  
}
```

```
package corejava;  
  
public class StringEndsWith {  
  
    public static void main(String[] args) {
```

```
String str = "Hello Java";

boolean result = str.endsWith("Java");

        System.out.println("String: " + str);

        System.out.println("Ends with Java: " + result);

    }

}

package corejava;

public class StringEquals {

    public static void main(String[] args) {

        String str1 = "Pujitha";

        String str2 = "Pujitha";

        boolean result = str1.equals(str2);

            System.out.println("First String: " + str1);

            System.out.println("Second String: " + str2);

            System.out.println("Strings are equal: " + result);

        }

    }

package corejava;

public class StringIndexOf;

    public static void main(String[] args) {

        String str = "Sai Pujitha";

        int result = str.indexOf('P');

            System.out.println("String: " + str);

            System.out.println("Index of P: " + result);

        }

    }
```

```
    }  
  
package corejava;  
  
public class StringIsEmpty {  
    public static void main(String[] args) {  
        String str =  
        boolean result = str.isEmpty();  
        System.out.println("String is empty: " + result);  
    }  
}
```

```
package corejava;  
  
public class StringLastIndexOf {  
    public static void main(String[] args) {  
        String str = "Hello Java";  
        int result = str.lastIndexOf('a');  
        System.out.println("String: " + str);  
        System.out.println("Last index of a: " + result);  
    }  
}
```

```
package corejava;  
  
public class StringLength {  
    public static void main(String[] args) {  
        String str = "Good Morning";  
        int n = str.length();  
        System.out.println("String: " + str);  
        System.out.println("Length of string: " + n);  
    }  
}
```

```

        }
    }

package corejava;

public class StringReplace {

    public static void main(String[] args) {

        String str = "Hello Java";

        String result = str.replace("Java", "World");

        System.out.println("Original String: " + str);

        System.out.println("After Replace: " + result);

    }

}

package corejava;

public class StringStartsWith {

    public static void main(String[] args) {

        String str = "Hello Java";

        boolean result = str.startsWith("Hello");

        System.out.println("String: " + str);

        System.out.println("Starts with Hello: " + result);

    }

}

package corejava;

public class StringSubString {

    public static void main(String[] args) {

        String str = "Hello Java";

        String result = str.substring(0, 5);
    }
}

```

```

        System.out.println("Original String: " + str);
        System.out.println("Substring: " + result);
    }
}

package corejava;

public class StringToLower {
    public static void main(String[] args) {
        String str = "HELLO WORLD";
        String result = str.toLowerCase();

        System.out.println("Original String: " + str);
        System.out.println("Lowercase String: " + result);
    }
}

package corejava;

public class StringToUpper {
    public static void main(String[] args) {
        String str = "My Collection";
        String result = str.toUpperCase();

        System.out.println("Original String: " + str);
        System.out.println("Uppercase String: " + result);
    }
}

package corejava;

public class StringTrim {
    public static void main(String[] args) ;

```

```

String str = "    Hello Java    ";
String result = str.trim();

        System.out.println("Original String: " + str);

        System.out.println("After trim: " + result);

    }

}

```

Output:

```

String: Hello Java
    Character at index 6: J

```

First String: Sai

Second String: Pujitha

Concatenated String: Sai Pujitha

```

String: Hello Java
Contains Java: true
String: Hello Java
Ends with Java: true

```

First String: Pujitha

Second String: Pujitha

Strings are equal: true

String: Sai Pujitha

Index of P: 4

String is empty: true

String: Hello Java

Last index of a: 9

String: Good Morning

Length of string: 12

Original String: Hello Java

After Replace: Hello World

String: Hello Java

Starts with Hello: true

Original String: Hello Java

Substring: Hello

Original String: HELLO WORLD

Lowercase String: hello world

Original String: My Collection

Uppercase String: MY COLLECTION

Original String: Hello Java

After trim: Hello Java

